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An exact algorithm for the identical parallel machine scheduling problem

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Abstract

The *NP-hard* classical deterministic scheduling problem of minimizing the makespan on identical parallel machines is considered. The polyhedral structure of the problem is analyzed, and strong valid inequalities are identified for fixed values of the makespan. Thus, partial linear descriptions of the associated polytope are obtained and used to build an exact algorithm. Linear programming relaxations are solved iteratively until the integer solution is reached. The algorithm includes a preprocessing phase which reformulates the problem and simplifies it. Different constructive rules have been tried out in the computation of upper bound values and the incidence of this choice is reported. Computational results show that the proposed algorithm yields, in a short computational time, optimal solutions in almost all tested cases.
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Keywords: Optimization; Polyhedral combinatorics; Scheduling

1. Introduction

In the classical deterministic identical parallel machine problem, there are a number of independent jobs to be processed on a range of identical machines. Each job has to be carried out on one of the machines during a fixed processing time, without preemption. The problem of finding the schedule that optimizes the makespan is considered.

Where can this model be applied to real-life situations? There are many examples in industry, social sciences and computer science, where the problem of job-allocation and total completion time fit in with this model: production lines, patients in hospitals, share-time systems, to give but three examples. In fact, any situation that deals with a series of processing units and has a set of jobs to be carried out on them are real-life examples of this mathematical model.

This problem is known to be *NP-complete* since PARTITION polynomially reduces to the special case with two machines [5]. So over the years there has been a great deal of research to develop approximation algorithms which can be applied. The only way to secure optimal solutions is by enumerative techniques. Dynamic programming and branch and bound (B&B) techniques have been

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Table 6
Performance of PA and B&B for larger size instances, $p_{ij} \in$ uniform in range [1–100]. This table gives average performance values over 10 problems

m	n	PA			B&B		
		Iterations	Time	Unsolved inst.	Iterations	Time	Unsolved inst.
3	20	25.40	0.12	0	3826073.50	1729.20	8
3	50	5.80	0.10	0	1451551.50	1203.65	6
3	100	17.00	0.22	0	777414.30	1210.26	6
3	200	4.60	0.12	0	133795.50	638.43	3
3	500	0.00	0.01	0	385.90	15.43	0
5	20	344.80	0.57	0	560053.90	431.09	2
5	50	29.20	0.35	0	1849177.30	1617.32	8
5	100	44.40	0.67	0	879344.70	1802.66	9
5	200	34.80	1.23	0	326452.70	1643.67	8
5	500	0.00	0.01	0	77960.20	1050.35	5
15	20	0.00	0.01	0	498004.30	918.36	4
15	50	103490.40	864.47	2	–	–	10
15	100	40153.00	329.66	0	–	–	10
15	200	11824.60	348.79	0	37489.60	1964.59	9
15	500	320.60	83.54	0	6411.60	1847.71	9
100	200	27.00	2042.08	0	–	–	10
100	500	292.20	2768.05	2	–	–	10
100	1000	182.00	5098.40	0	–	–	10

instances like 5–15, 10–25 and 15–50, while MF_{17} is known to be more efficient solving easy instances.

To analyse the performance of PA, we have compared it with the B&B algorithm which forms part of the CPLEX Callable Library 6.5. We have considered three classes of test problems obtained by randomly generating the p_i values according to uniform distributions in the ranges [1–100], [10–100] and [50–100]. The upper bound for PA has been computed by LPT in this experiment. For each class, and for different values of n and m, the entries in Tables 3–5 give the average number of iterations and of CPU time in seconds, required by each of the two algorithms, computed over 10 problem instances. For PA the number of iterations is the sum of the number of solved LP relaxations and of the nodes of the B&B procedure (when PA makes use of B&B to complete the search). For the B&B an iteration is every solved node.

The computational performance of PA was in general satisfactory for all tested cases. Given the high computational complexity of the problem in hand PA and B&B could stop without guaranteeing that the obtained solution is optimal. In the tested cases, that we present in Tables 3–5, both were always able to converge to the optimal integral solution. Table 6 presents the performance of the PA and B&B for larger size problems. It is important to notice that the B&B was not capable of optimizing a high percentage of generated instances (the number of unsolved instances is indicated when it is greater than zero). Furthermore, we can see that there is an important saving in the average computational effort of the PA.

As expected, the PA reaches the optimal solution in many more cases for the same capacity of memory and in less time when we apply it to $P//C_{\max}$ than when we apply it to $R//C_{\max}$. Therefore application to $P//C_{\max}$ is even more efficient than the results presented in [9].

5. Concluding remarks

In this paper we propose a new algorithm based on cutting plane techniques for a hard scheduling problem. We have followed the trend of recent CO literature, which applies Polyhedral Theory to optimize NP-hard combinatorial problems.

The obtained results show that this kind of approach is quite a powerful tool for effectively producing quality feasible solutions. They also give support to the hypothesis stating that specially-developed algorithms for specific combinatorial problems work better than general methods like B&B.

After having applied this technique to $P//C_{\max}$ and $R//C_{\max}$, and having refined the computation of initial lower and upper bound, we have been able to prove the efficiency of this algorithm and thus we are now considering the development of similar approaches to solve even harder scheduling problems such as job-shop problems.

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